CORE PURE / VECTORS, MATRICES & LINEAR TRANSFORMATIONS

Vector Product

Worked solutions 10 questions · 83 marks

Questions

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1. **(a)** The vectors

$$\mathbf{p} = \begin{pmatrix} 1 \\ 3 \\ 2 \end{pmatrix} \quad \text{and} \quad \mathbf{q} = \begin{pmatrix} 2 \\ m \\ -1 \end{pmatrix}$$

span a plane with normal vector

$$\mathbf{n} = \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}$$

Given that $\mathbf{q} \times \mathbf{p}$ is in the same direction as $\mathbf{n}$, find m and the positive constant λ such that

$$\mathbf{q} \times \mathbf{p} = \lambda \mathbf{n}$$ **[3]**

(b) Using $\mathbf{q} \times \mathbf{p}$, determine the acute angle between $\mathbf{p}$ and $\mathbf{q}$. **[3]**

Solution

(a) Taking

$$\mathbf{q} = \begin{pmatrix} 2 \\ m \\ -1 \end{pmatrix}, \qquad \mathbf{p} = \begin{pmatrix} 1 \\ 3 \\ 2 \end{pmatrix}$$

we find the cross product:

$$\begin{aligned} \mathbf{q} \times \mathbf{p} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & m & -1 \\ 1 & 3 & 2 \end{vmatrix} \\ &= \mathbf{i}(2m+3) - \mathbf{j}(4+1) + \mathbf{k}(6-m) \\ &= \begin{pmatrix} 2m+3 \\ -5 \\ 6-m \end{pmatrix} \end{aligned}$$

Since $\mathbf{q} \times \mathbf{p}$ is in the same direction as

$$\mathbf{n} = \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}$$

we can write

$$\mathbf{q} \times \mathbf{p} = \lambda \mathbf{n} = \begin{pmatrix} \lambda \\ -\lambda \\ \lambda \end{pmatrix}$$

So

$$\begin{pmatrix} 2m+3 \\ -5 \\ 6-m \end{pmatrix} = \begin{pmatrix} \lambda \\ -\lambda \\ \lambda \end{pmatrix}$$

Equating components:

$$-5 = -\lambda \implies \lambda = 5$$

Then

$$2m + 3 = 5 \implies 2m = 2 \implies m = 1$$

Checking the third component:

$$6 - m = 6 - 1 = 5$$

which is consistent.

Hence

$$m = 1, \qquad \lambda = 5$$

(b) Using $m = 1$,

$$\mathbf{q} = \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix}$$

Also

$$\mathbf{q} \times \mathbf{p} = \begin{pmatrix} 5 \\ -5 \\ 5 \end{pmatrix}$$

so

$$\begin{aligned} |\mathbf{q} \times \mathbf{p}| &= \sqrt{5^2 + (-5)^2 + 5^2} \\ &= \sqrt{75} \\ &= 5\sqrt{3} \end{aligned}$$

Now

$$\begin{aligned} |\mathbf{p}| &= \sqrt{1^2 + 3^2 + 2^2} = \sqrt{14} \\ |\mathbf{q}| &= \sqrt{2^2 + 1^2 + (-1)^2} = \sqrt{6} \end{aligned}$$

Using

$$|\mathbf{q} \times \mathbf{p}| = |\mathbf{q}|\,|\mathbf{p}| \sin\theta$$

gives

$$\begin{aligned} \sin\theta &= \frac{|\mathbf{q} \times \mathbf{p}|}{|\mathbf{q}|\,|\mathbf{p}|} \\ &= \frac{5\sqrt{3}}{\sqrt{6}\,\sqrt{14}} \\ &= \frac{5\sqrt{3}}{\sqrt{84}} \\ &= \frac{5}{2\sqrt{7}} \end{aligned}$$

So the acute angle is

$$\theta = \sin^{-1}\left(\frac{5}{2\sqrt{7}}\right) \approx 70.9^\circ$$

2. You are given the fixed points $A(1,0,1)$ and $C(4,0,2)$, and the variable point $B(t,2,5-t)$, where t is a real parameter.

(a) Using only the geometrical properties of the vector product, explain why the statement "$\overrightarrow{AB}\times\overrightarrow{AC}=\mathbf{0}$" is false for all values of t. **[2]**

(b) (i) Use the vector product to find an expression, in terms of t, for the area of triangle ABC. **[4]**
(ii) Hence determine the value of t for which the area of triangle ABC is a minimum. **[2]**

Solution

(a)

$$\overrightarrow{AB}=\begin{pmatrix}t-1\\2\\4-t\end{pmatrix},\qquad \overrightarrow{AC}=\begin{pmatrix}3\\0\\1\end{pmatrix}$$

Geometrically, $\overrightarrow{AB}\times\overrightarrow{AC}=\mathbf{0}$ only if $\overrightarrow{AB}$ and $\overrightarrow{AC}$ are parallel, or if one of them is the zero vector.
Here $\overrightarrow{AC}\neq\mathbf{0}$, and $\overrightarrow{AB}$ can never be parallel to $\overrightarrow{AC}$ because:

$$y\text{-component of }\overrightarrow{AB}=2,\qquad y\text{-component of }\overrightarrow{AC}=0$$

So $\overrightarrow{AB}$ is not a scalar multiple of $\overrightarrow{AC}$ for any real t.
Therefore $\overrightarrow{AB}\times\overrightarrow{AC}\neq\mathbf{0}$ for all values of t, so the statement is false for all t.

(b) (i) First find the two sides of the triangle:

$$\overrightarrow{AB}=\begin{pmatrix}t-1\\2\\4-t\end{pmatrix},\qquad \overrightarrow{AC}=\begin{pmatrix}3\\0\\1\end{pmatrix}$$

Now calculate the vector product:

$$\begin{aligned}\overrightarrow{AB}\times\overrightarrow{AC}&=\begin{vmatrix}\mathbf{i}&\mathbf{j}&\mathbf{k}\\t-1&2&4-t\\3&0&1\end{vmatrix}\\&=\mathbf{i}\big(2\cdot1-(4-t)\cdot0\big)-\mathbf{j}\big((t-1)\cdot1-(4-t)\cdot3\big)+\mathbf{k}\big((t-1)\cdot0-2\cdot3\big)\\&=2\mathbf{i}-\mathbf{j}\big(t-1-12+3t\big)-6\mathbf{k}\\&=2\mathbf{i}-(4t-13)\mathbf{j}-6\mathbf{k}\\&=\begin{pmatrix}2\\13-4t\\-6\end{pmatrix}\end{aligned}$$

The area of triangle ABC is

$$\frac{1}{2}\left|\overrightarrow{AB}\times\overrightarrow{AC}\right|$$

So

$$\begin{aligned}\text{Area}&=\frac{1}{2}\sqrt{2^2+(13-4t)^2+(-6)^2}\\&=\frac{1}{2}\sqrt{4+(13-4t)^2+36}\\&=\frac{1}{2}\sqrt{(13-4t)^2+40}\end{aligned}$$

Hence the area of triangle ABC is

$$\frac{1}{2}\sqrt{(13-4t)^2+40}$$

(ii) To minimise the area, it is enough to minimise

$$(13-4t)^2+40$$

Since $(13-4t)^2 \geq 0$, its smallest value is 0, which occurs when

$$13-4t=0$$

Hence

$$t=\frac{13}{4}$$

The minimum area is then

$$\frac{1}{2}\sqrt{40}=\sqrt{10}$$

Therefore the area is minimised when $t=\frac{13}{4}$, and the minimum area is $\sqrt{10}$.

3. With respect to a fixed origin O, the points A and B have coordinates $(1, 1, 0)$ and $(p, 3, 2)$ respectively, where p is a constant.

For each of the following, determine the possible values of p for which

(a) $\overrightarrow{AB}$ makes an angle of $60°$ with the positive y-axis. **[3]**

(b) $\overrightarrow{OA} \times \overrightarrow{AB}$ is parallel to

$$\begin{pmatrix} 2 \\ -2 \\ 1 \end{pmatrix}$$ **[3]**

(c) the area of triangle OAB is $\frac{3}{2}$. **[3]**

Solution

(a) First find the vector

$$\overrightarrow{AB} = \begin{pmatrix} p-1 \\ 3-1 \\ 2-0 \end{pmatrix} = \begin{pmatrix} p-1 \\ 2 \\ 2 \end{pmatrix}$$

The positive y-axis has direction vector

$$\mathbf{j} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$$

Using the scalar product formula,

$$\overrightarrow{AB} \cdot \mathbf{j} = \left|\overrightarrow{AB}\right| |\mathbf{j}| \cos 60°$$

So

$$\begin{aligned} 2 &= \sqrt{(p-1)^2 + 2^2 + 2^2} \times 1 \times \frac{1}{2} \\ 2 &= \frac{1}{2}\sqrt{(p-1)^2 + 8} \\ 4 &= \sqrt{(p-1)^2 + 8} \\ 16 &= (p-1)^2 + 8 \\ (p-1)^2 &= 8 \end{aligned}$$

Hence

$$p - 1 = \pm 2\sqrt{2}$$

Therefore the possible values of p are

$$p = 1 \pm 2\sqrt{2}$$

(b)

$$\overrightarrow{OA} = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \qquad \overrightarrow{AB} = \begin{pmatrix} p-1 \\ 2 \\ 2 \end{pmatrix}$$

Now calculate the cross product:

$$\begin{aligned}\overrightarrow{OA} \times \overrightarrow{AB} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 1 & 0 \\ p-1 & 2 & 2 \end{vmatrix} \\ &= \mathbf{i}(1 \cdot 2 - 0 \cdot 2) - \mathbf{j}(1 \cdot 2 - 0 \cdot (p-1)) + \mathbf{k}(1 \cdot 2 - 1 \cdot (p-1)) \\ &= \begin{pmatrix} 2 \\ -2 \\ 3-p \end{pmatrix}\end{aligned}$$

For this to be parallel to $\begin{pmatrix} 2 \\ -2 \\ 1 \end{pmatrix}$, we must have

$$\begin{pmatrix} 2 \\ -2 \\ 3-p \end{pmatrix} = \lambda \begin{pmatrix} 2 \\ -2 \\ 1 \end{pmatrix}$$

Comparing first components,

$$2 = 2\lambda \implies \lambda = 1$$

Then comparing third components,

$$3 - p = 1$$

So

$$p = 2$$

(c) For triangle OAB, the area is

$$\text{Area} = \frac{1}{2}\left|\overrightarrow{OA} \times \overrightarrow{OB}\right|$$

Since

$$\overrightarrow{OB} = \overrightarrow{OA} + \overrightarrow{AB}$$

we have

$$\begin{aligned}\overrightarrow{OA} \times \overrightarrow{OB} &= \overrightarrow{OA} \times (\overrightarrow{OA} + \overrightarrow{AB}) \\ &= \overrightarrow{OA} \times \overrightarrow{OA} + \overrightarrow{OA} \times \overrightarrow{AB} \\ &= \mathbf{0} + \overrightarrow{OA} \times \overrightarrow{AB} \\ &= \begin{pmatrix} 2 \\ -2 \\ 3-p \end{pmatrix}\end{aligned}$$

Therefore

$$\begin{aligned}\frac{1}{2}\sqrt{2^2 + (-2)^2 + (3-p)^2} &= \frac{3}{2} \\ \frac{1}{2}\sqrt{8 + (3-p)^2} &= \frac{3}{2} \\ \sqrt{8 + (3-p)^2} &= 3 \\ 8 + (3-p)^2 &= 9 \\ (3-p)^2 &= 1\end{aligned}$$

So

$$3 - p = \pm 1$$

Hence the possible values of p are

$$p = 2 \text{ or } p = 4$$

4. **(a)** Let

$$\mathbf{p} = \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix} \quad \text{and} \quad \mathbf{q} = \begin{pmatrix} x \\ 0 \\ 1 \end{pmatrix}$$

The area of the parallelogram formed by $\mathbf{p}$ and $\mathbf{q}$ is $\sqrt{2}$.
It is also given that $\mathbf{p} \times \mathbf{q}$ is perpendicular to

$$\begin{pmatrix} 1 \\ 3 \\ m \end{pmatrix}$$

Find x and m. **[4]**

(b) Hence, using $|\mathbf{p} \times \mathbf{q}|$, determine the acute angle between $\mathbf{p}$ and $\mathbf{q}$. **[3]**

Solution

(a) For two vectors, the area of the parallelogram they form is

$$|\mathbf{p} \times \mathbf{q}|$$

First find $\mathbf{p} \times \mathbf{q}$:

$$\begin{aligned} \mathbf{p} \times \mathbf{q} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 1 & 2 \\ x & 0 & 1 \end{vmatrix} \\ &= \mathbf{i}(1 \cdot 1 - 2 \cdot 0) - \mathbf{j}(1 \cdot 1 - 2x) + \mathbf{k}(1 \cdot 0 - 1 \cdot x) \\ &= \mathbf{i} - \mathbf{j}(1 - 2x) - x\mathbf{k} \\ &= \begin{pmatrix} 1 \\ 2x - 1 \\ -x \end{pmatrix} \end{aligned}$$

Since the area is $\sqrt{2}$,

$$\begin{aligned} |\mathbf{p} \times \mathbf{q}| &= \sqrt{2} \\ \therefore \ |\mathbf{p} \times \mathbf{q}|^2 &= 2 \end{aligned}$$

So

$$\begin{aligned} 1^2 + (2x - 1)^2 + (-x)^2 &= 2 \\ 1 + (4x^2 - 4x + 1) + x^2 &= 2 \\ 5x^2 - 4x &= 0 \\ x(5x - 4) &= 0 \end{aligned}$$

Hence

$$x = 0 \quad \text{or} \quad x = \frac{4}{5}$$

Now use the condition that $\mathbf{p} \times \mathbf{q}$ is perpendicular to $\begin{pmatrix} 1 \\ 3 \\ m \end{pmatrix}$.

Perpendicular vectors have zero dot product, so

$$\begin{aligned} \begin{pmatrix} 1 \\ 2x - 1 \\ -x \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 3 \\ m \end{pmatrix} &= 0 \\ 1 + 3(2x - 1) + (-x)m &= 0 \\ 1 + 6x - 3 - mx &= 0 \\ x(6 - m) &= 2 \end{aligned}$$

This shows $x \neq 0$, so the value $x = 0$ must be rejected.

Therefore

$$x = \frac{4}{5}$$

Substitute into $x(6 - m) = 2$:

$$\frac{4}{5}(6 - m) = 2$$
$$6 - m = \frac{5}{2}$$
$$m = 6 - \frac{5}{2} = \frac{7}{2}$$

Therefore,

$$x = \frac{4}{5}, \qquad m = \frac{7}{2}$$

(b) Using

$$|\mathbf{p} \times \mathbf{q}| = |\mathbf{p}|\,|\mathbf{q}| \sin\theta$$

where θ is the angle between $\mathbf{p}$ and $\mathbf{q}$.

From part (a),

$$|\mathbf{p} \times \mathbf{q}| = \sqrt{2}$$

Also,

$$|\mathbf{p}| = \sqrt{1^2 + 1^2 + 2^2} = \sqrt{6}$$

and with $x = \frac{4}{5}$,

$$|\mathbf{q}| = \sqrt{\left(\frac{4}{5}\right)^2 + 0^2 + 1^2}$$
$$= \sqrt{\frac{16}{25} + 1} = \sqrt{\frac{41}{25}} = \frac{\sqrt{41}}{5}$$

So

$$\sqrt{2} = \sqrt{6} \cdot \frac{\sqrt{41}}{5} \sin\theta$$
$$\sin\theta = \frac{\sqrt{2}}{\sqrt{6}(\sqrt{41}/5)}$$
$$= \frac{5\sqrt{2}}{\sqrt{246}}$$
$$= \frac{5}{\sqrt{123}}$$

Hence the acute angle is

$$\theta = \sin^{-1}\left(\frac{5}{\sqrt{123}}\right) \approx 26.8^\circ$$

5. The point P has coordinates $P(2,-1,1)$ and the midpoint of PR is $M(1,0,2)$. The vector $\overrightarrow{PQ}$ is given by

$$\overrightarrow{PQ} = \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}$$

(a) Use a vector product to show that the area of triangle PQR is $\sqrt{14}$. **[4]**

(b) Find a vector equation of the plane containing P, Q and R in the form $\mathbf{r} \cdot \mathbf{n} = k$. **[1]**

(c) Hence find the exact distance of the plane from the origin. **[1]**

Solution

(a) First find the coordinates of Q and R.

Since $\overrightarrow{OQ} = \overrightarrow{OP} + \overrightarrow{PQ}$,

$$\overrightarrow{OQ} = \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} + \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} = \begin{pmatrix} 3 \\ 1 \\ 1 \end{pmatrix}$$

so Q is the point $(3,1,1)$.

Since $M(1,0,2)$ is the midpoint of PR,

$$\overrightarrow{OM} = \frac{1}{2}\left(\overrightarrow{OP} + \overrightarrow{OR}\right)$$

so

$$\overrightarrow{OR} = 2\overrightarrow{OM} - \overrightarrow{OP} = \begin{pmatrix} 2 \\ 0 \\ 4 \end{pmatrix} - \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 3 \end{pmatrix}$$

Now

$$\overrightarrow{PR} = \overrightarrow{OR} - \overrightarrow{OP} = \begin{pmatrix} 0 \\ 1 \\ 3 \end{pmatrix} - \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} = \begin{pmatrix} -2 \\ 2 \\ 2 \end{pmatrix}$$

Using the vector product,

$$\overrightarrow{PQ} \times \overrightarrow{PR} = \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} \times \begin{pmatrix} -2 \\ 2 \\ 2 \end{pmatrix} = \begin{pmatrix} 2 \cdot 2 - 0 \cdot 2 \\ 0 \cdot (-2) - 1 \cdot 2 \\ 1 \cdot 2 - 2 \cdot (-2) \end{pmatrix} = \begin{pmatrix} 4 \\ -2 \\ 6 \end{pmatrix}$$

Its magnitude is

$$\left|\overrightarrow{PQ} \times \overrightarrow{PR}\right| = \sqrt{4^2 + (-2)^2 + 6^2} = \sqrt{16 + 4 + 36} = \sqrt{56} = 2\sqrt{14}$$

The area of triangle PQR is half the area of the parallelogram:

$$\text{Area of } \triangle PQR = \frac{1}{2}\left|\overrightarrow{PQ} \times \overrightarrow{PR}\right| = \frac{1}{2}(2\sqrt{14}) = \sqrt{14}$$

Hence the area of triangle PQR is $\sqrt{14}$.

(b) A normal vector to the plane is

$$\mathbf{n} = \begin{pmatrix} 4 \\ -2 \\ 6 \end{pmatrix}$$

which can be simplified to

$$\mathbf{n} = \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix}$$

Using the point $P(2, -1, 1)$,

$$k = \mathbf{p} \cdot \mathbf{n} = \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix} = 4 + 1 + 3 = 8$$

So the plane is

$$\mathbf{r} \cdot \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix} = 8$$

(c) For a plane $\mathbf{r} \cdot \mathbf{n} = d$, the distance from a point $\mathbf{p}$ is

$$\text{dist} = \frac{|\mathbf{p} \cdot \mathbf{n} - d|}{|\mathbf{n}|}$$

Here the point is the origin, so $\mathbf{p} = \mathbf{0}$, and the plane is

$$\mathbf{r} \cdot \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix} = 8$$

Therefore

$$\text{dist} = \frac{|0 - 8|}{\sqrt{2^2 + (-1)^2 + 3^2}} = \frac{8}{\sqrt{14}} = \frac{4\sqrt{14}}{7}$$

Hence the exact distance of the plane from the origin is $\frac{4\sqrt{14}}{7}$.

6. The points P, Q and R have position vectors $\mathbf{p} = 2\mathbf{i} - 2\mathbf{k}$, $\mathbf{q} = 3\mathbf{j} + 3\mathbf{k}$ and $\mathbf{r} = t\mathbf{i} - \mathbf{j} + \mathbf{k}$ respectively, relative to the origin O.

Determine the value(s) of t in each of the following cases.

(a) The line OR is parallel to a normal to the plane OPQ. **[2]**

(b) The volume of tetrahedron $OPQR$ is 9. **[4]**

Solution

(a) The plane OPQ contains the vectors

$$\mathbf{p} = \begin{pmatrix} 2 \\ 0 \\ -2 \end{pmatrix}, \qquad \mathbf{q} = \begin{pmatrix} 0 \\ 3 \\ 3 \end{pmatrix}$$

A normal to the plane is $\mathbf{p} \times \mathbf{q}$:

$$\begin{aligned} \mathbf{p} \times \mathbf{q} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & 0 & -2 \\ 0 & 3 & 3 \end{vmatrix} \\ &= \mathbf{i}(0 \cdot 3 - (-2) \cdot 3) - \mathbf{j}(2 \cdot 3 - (-2) \cdot 0) + \mathbf{k}(2 \cdot 3 - 0 \cdot 0) \\ &= 6\mathbf{i} - 6\mathbf{j} + 6\mathbf{k} \\ &= 6(\mathbf{i} - \mathbf{j} + \mathbf{k}) \end{aligned}$$

The line OR has direction vector

$$\mathbf{r} = t\mathbf{i} - \mathbf{j} + \mathbf{k}$$

If OR is parallel to a normal to the plane, then $\mathbf{r}$ must be parallel to $\mathbf{i} - \mathbf{j} + \mathbf{k}$.

So

$$\begin{pmatrix} t \\ -1 \\ 1 \end{pmatrix} \parallel \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}$$

Comparing components gives $t = 1$.

(b) Using the vector formula for the volume of a tetrahedron,

$$\text{Volume of } OPQR = \frac{1}{6} \left| \mathbf{p} \cdot (\mathbf{q} \times \mathbf{r}) \right|$$

Equivalently,

$$\text{Volume} = \frac{1}{6} \left| (\mathbf{p} \times \mathbf{q}) \cdot \mathbf{r} \right|$$

From part (a),

$$\mathbf{p} \times \mathbf{q} = 6(\mathbf{i} - \mathbf{j} + \mathbf{k})$$

Also,

$$\mathbf{r} = t\mathbf{i} - \mathbf{j} + \mathbf{k}$$

So the scalar triple product is

$$\begin{aligned} (\mathbf{p} \times \mathbf{q}) \cdot \mathbf{r} &= 6(\mathbf{i} - \mathbf{j} + \mathbf{k}) \cdot (t\mathbf{i} - \mathbf{j} + \mathbf{k}) \\ &= 6(t + 1 + 1) \\ &= 6(t + 2) \end{aligned}$$

Therefore

$$\begin{aligned}\text{Volume} &= \frac{1}{6}|6(t+2)| \\ &= |t+2|\end{aligned}$$

Given that the volume is 9,

$$|t+2| = 9$$

So

$$\begin{aligned} t+2 = 9 \quad &\text{or} \quad t+2 = -9 \\ t = 7 \quad &\text{or} \quad t = -11 \end{aligned}$$

Hence the values of t are 7 and -11.

7. Points P, Q and R have position vectors $\mathbf{p}$, $\mathbf{q}$ and $\mathbf{r}$ respectively, relative to origin O.

It is given that $(\mathbf{q}-\mathbf{p})\times(\mathbf{r}-\mathbf{q})=2\mathbf{p}$ and that $|\mathbf{p}|=5$.

(a) Determine each of the following as either a single vector or a scalar quantity.

(i) $(\mathbf{r}-\mathbf{q})\times(\mathbf{q}-\mathbf{p})$ **[1]**

(ii) $\mathbf{p}\times\big((\mathbf{q}-\mathbf{p})\times(\mathbf{r}-\mathbf{q})\big)$ **[2]**

(iii) $\mathbf{p}\cdot\big((\mathbf{q}-\mathbf{p})\times(\mathbf{r}-\mathbf{q})\big)$ **[2]**

(b) Describe a geometrical relationship between the points O, P, Q and R which can be deduced from

(i) the statement $(\mathbf{q}-\mathbf{p})\times(\mathbf{r}-\mathbf{q})=2\mathbf{p}$ **[1]**

(ii) the result of part (a)(iii) **[1]**

Solution

(a) (i) Using the fact that $\mathbf{a}\times\mathbf{b}=-\mathbf{b}\times\mathbf{a}$,

$$(\mathbf{r}-\mathbf{q})\times(\mathbf{q}-\mathbf{p})=-\big((\mathbf{q}-\mathbf{p})\times(\mathbf{r}-\mathbf{q})\big)=-2\mathbf{p}$$

So the vector is $-2\mathbf{p}$.

(ii) Substitute the given result directly:

$$\mathbf{p}\times\big((\mathbf{q}-\mathbf{p})\times(\mathbf{r}-\mathbf{q})\big)=\mathbf{p}\times2\mathbf{p}=2(\mathbf{p}\times\mathbf{p})=\mathbf{0}$$

So the vector is $\mathbf{0}$.

(iii) Again using the given result,

$$\begin{aligned}\mathbf{p}\cdot\big((\mathbf{q}-\mathbf{p})\times(\mathbf{r}-\mathbf{q})\big)&=\mathbf{p}\cdot2\mathbf{p}\\&=2(\mathbf{p}\cdot\mathbf{p})\\&=2|\mathbf{p}|^2\\&=2(5^2)\\&=50\end{aligned}$$

So the scalar is 50.

(b) (i) The vector $\mathbf{q}-\mathbf{p}$ is in the direction of PQ, and $\mathbf{r}-\mathbf{q}$ is in the direction of QR.
Their cross product is perpendicular to both of these vectors. Since

$$(\mathbf{q}-\mathbf{p})\times(\mathbf{r}-\mathbf{q})=2\mathbf{p}$$

the vector $\mathbf{p}$ is parallel to this perpendicular direction.
But $\mathbf{p}$ is the position vector of P, so it is the direction of OP. Therefore OP is perpendicular to both PQ and QR, so OP is normal to the plane PQR.
Hence $OP\perp$ plane PQR.

(ii) From part (a)(iii),

$$\mathbf{p}\cdot\big((\mathbf{q}-\mathbf{p})\times(\mathbf{r}-\mathbf{q})\big)=50\neq0$$

A non-zero scalar triple product means the three vectors involved are not coplanar.
Therefore the points O, P, Q and R are not coplanar.

8. The points A, B and C lie on the plane Π.

The position vectors of A and B are $\mathbf{a} = 2\mathbf{i} - \mathbf{j} + 3\mathbf{k}$ and $\mathbf{b} = 5\mathbf{i} + \mathbf{j}$ respectively.

The centroid G of triangle ABC has position vector $4\mathbf{i} + \mathbf{j} + 2\mathbf{k}$.

(a) Find $\overrightarrow{AB} \times \overrightarrow{AC}$. **[4]**

(b) Find an equation for Π in the form $\mathbf{r} \cdot \mathbf{n} = p$. **[2]**

The point D has position vector $\mathbf{i} + 5\mathbf{j} + 4\mathbf{k}$.

(c) Determine the volume of the tetrahedron $ABCD$. **[4]**

Solution

(a) The centroid of triangle ABC has position vector

$$\mathbf{g} = \frac{\mathbf{a} + \mathbf{b} + \mathbf{c}}{3}$$

So

$$\mathbf{c} = 3\mathbf{g} - \mathbf{a} - \mathbf{b}$$

Using

$$\mathbf{a} = \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 5 \\ 1 \\ 0 \end{pmatrix}, \quad \mathbf{g} = \begin{pmatrix} 4 \\ 1 \\ 2 \end{pmatrix}$$

gives

$$\begin{aligned} \mathbf{c} &= 3\begin{pmatrix} 4 \\ 1 \\ 2 \end{pmatrix} - \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix} - \begin{pmatrix} 5 \\ 1 \\ 0 \end{pmatrix} \\ &= \begin{pmatrix} 12 \\ 3 \\ 6 \end{pmatrix} - \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix} - \begin{pmatrix} 5 \\ 1 \\ 0 \end{pmatrix} \\ &= \begin{pmatrix} 5 \\ 3 \\ 3 \end{pmatrix} \end{aligned}$$

Now

$$\overrightarrow{AB} = \mathbf{b} - \mathbf{a} = \begin{pmatrix} 5 \\ 1 \\ 0 \end{pmatrix} - \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix} = \begin{pmatrix} 3 \\ 2 \\ -3 \end{pmatrix}$$

$$\overrightarrow{AC} = \mathbf{c} - \mathbf{a} = \begin{pmatrix} 5 \\ 3 \\ 3 \end{pmatrix} - \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix} = \begin{pmatrix} 3 \\ 4 \\ 0 \end{pmatrix}$$

Hence

$$\begin{aligned} \overrightarrow{AB} \times \overrightarrow{AC} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 3 & 2 & -3 \\ 3 & 4 & 0 \end{vmatrix} \\ &= \mathbf{i}(2 \cdot 0 - (-3) \cdot 4) - \mathbf{j}(3 \cdot 0 - (-3) \cdot 3) + \mathbf{k}(3 \cdot 4 - 2 \cdot 3) \\ &= 12\mathbf{i} - 9\mathbf{j} + 6\mathbf{k} \end{aligned}$$

(b) A normal vector to the plane is $\overrightarrow{AB} \times \overrightarrow{AC}$, so we may take

$$\mathbf{n} = 12\mathbf{i} - 9\mathbf{j} + 6\mathbf{k} = 3(4\mathbf{i} - 3\mathbf{j} + 2\mathbf{k})$$

A simpler normal vector is

$$\mathbf{n} = 4\mathbf{i} - 3\mathbf{j} + 2\mathbf{k}$$

The plane has equation $\mathbf{r} \cdot \mathbf{n} = p$. Since A lies on the plane,

$$\begin{aligned} p &= \mathbf{a} \cdot \mathbf{n} \\ &= \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix} \cdot \begin{pmatrix} 4 \\ -3 \\ 2 \end{pmatrix} \\ &= 2(4) + (-1)(-3) + 3(2) \\ &= 8 + 3 + 6 = 17 \end{aligned}$$

Therefore the equation of Π is

$$\mathbf{r} \cdot (4\mathbf{i} - 3\mathbf{j} + 2\mathbf{k}) = 17$$

(c) First find

$$\mathbf{d} = \begin{pmatrix} 1 \\ 5 \\ 4 \end{pmatrix}$$

so

$$\overrightarrow{AD} = \mathbf{d} - \mathbf{a} = \begin{pmatrix} 1 \\ 5 \\ 4 \end{pmatrix} - \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix} = \begin{pmatrix} -1 \\ 6 \\ 1 \end{pmatrix}$$

The volume of a tetrahedron with three edges $\overrightarrow{AB}$, $\overrightarrow{AC}$, $\overrightarrow{AD}$ meeting at A is

$$V = \frac{1}{6} \left| (\overrightarrow{AB} \times \overrightarrow{AC}) \cdot \overrightarrow{AD} \right|$$

Using the result from part (a),

$$\begin{aligned} V &= \frac{1}{6} \left| \begin{pmatrix} 12 \\ -9 \\ 6 \end{pmatrix} \cdot \begin{pmatrix} -1 \\ 6 \\ 1 \end{pmatrix} \right| \\ &= \frac{1}{6} |12(-1) + (-9)(6) + 6(1)| \\ &= \frac{1}{6} |-12 - 54 + 6| \\ &= \frac{1}{6} |-60| \\ &= 10 \end{aligned}$$

So the volume of tetrahedron $ABCD$ is 10.

9. The lines L_1, L_2 and L_3 are given by

$$L_1: \quad \mathbf{r} = \begin{pmatrix} 8 \\ 3 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ 2 \\ -1 \end{pmatrix}$$

$$L_2: \quad x = 9 - 4\mu,\ y = -2 + 3\mu,\ z = -1 + 3\mu$$

$$L_3: \quad \frac{x-3}{1} = \frac{y+6}{-5} = \frac{z-1}{-2}$$

The pairwise intersections of these lines are the vertices of a triangle.

Find the area of this triangle. **[9]**

Solution

Write the lines in parametric form:

$$L_1: \ (x, y, z) = (8 + 3\lambda,\ 3 + 2\lambda,\ 1 - \lambda)$$

$$L_2: \ (x, y, z) = (9 - 4\mu,\ -2 + 3\mu,\ -1 + 3\mu)$$

and for L_3, let the common value be t. Then

$$\frac{x-3}{1} = \frac{y+6}{-5} = \frac{z-1}{-2} = t$$

so

$$L_3: \ (x, y, z) = (3 + t,\ -6 - 5t,\ 1 - 2t)$$

Let the three vertices of the triangle be

$$A = L_1 \cap L_2, \qquad B = L_1 \cap L_3, \qquad C = L_2 \cap L_3$$

For $A = L_1 \cap L_2$, equate coordinates:

$$\begin{aligned} 8 + 3\lambda &= 9 - 4\mu \\ 3 + 2\lambda &= -2 + 3\mu \\ 1 - \lambda &= -1 + 3\mu \end{aligned}$$

From the third equation,

$$2 - \lambda = 3\mu$$

so

$$\mu = \frac{2 - \lambda}{3}$$

Substitute into the first equation:

$$\begin{aligned} 8 + 3\lambda &= 9 - \frac{4(2-\lambda)}{3} \\ 24 + 9\lambda &= 27 - 8 + 4\lambda \\ 24 + 9\lambda &= 19 + 4\lambda \\ 5\lambda &= -5 \\ \lambda &= -1 \end{aligned}$$

Hence

$$\mu = \frac{2 - (-1)}{3} = 1$$

So

$$A = (8 + 3(-1),\ 3 + 2(-1),\ 1 - (-1)) = (5, 1, 2)$$

For $B = L_1 \cap L_3$, equate coordinates:

$$\begin{aligned} 8 + 3\lambda &= 3 + t \\ 3 + 2\lambda &= -6 - 5t \\ 1 - \lambda &= 1 - 2t \end{aligned}$$

From the third equation,

$$\lambda = 2t$$

Substitute into the first equation:

$$\begin{aligned} 8 + 3(2t) &= 3 + t \\ 8 + 6t &= 3 + t \\ 5t &= -5 \\ t &= -1 \end{aligned}$$

Hence

$$\lambda = 2(-1) = -2$$

So

$$B = (8 + 3(-2),\ 3 + 2(-2),\ 1 - (-2)) = (2, -1, 3)$$

For $C = L_2 \cap L_3$, equate coordinates:

$$\begin{aligned} 9 - 4\mu &= 3 + t \\ -2 + 3\mu &= -6 - 5t \\ -1 + 3\mu &= 1 - 2t \end{aligned}$$

From the first equation,

$$t = 6 - 4\mu$$

Substitute into the third equation:

$$\begin{aligned} -1 + 3\mu &= 1 - 2(6 - 4\mu) \\ -1 + 3\mu &= 1 - 12 + 8\mu \\ -1 + 3\mu &= -11 + 8\mu \\ 10 &= 5\mu \\ \mu &= 2 \end{aligned}$$

Hence

$$t = 6 - 4(2) = -2$$

So

$$C = (9 - 4(2),\ -2 + 3(2),\ -1 + 3(2)) = (1, 4, 5)$$

Now use the formula for the area of a triangle:

$$\text{Area} = \frac{1}{2} |\mathbf{u} \times \mathbf{v}|$$

where $\mathbf{u}$ and $\mathbf{v}$ are two side vectors.
Take

$$\overrightarrow{AB} = \begin{pmatrix} 2-5 \\ -1-1 \\ 3-2 \end{pmatrix} = \begin{pmatrix} -3 \\ -2 \\ 1 \end{pmatrix}, \qquad \overrightarrow{AC} = \begin{pmatrix} 1-5 \\ 4-1 \\ 5-2 \end{pmatrix} = \begin{pmatrix} -4 \\ 3 \\ 3 \end{pmatrix}$$

Then

$$\overrightarrow{AB} \times \overrightarrow{AC} = \begin{pmatrix} (-2)(3) - (1)(3) \\ (1)(-4) - (-3)(3) \\ (-3)(3) - (-2)(-4) \end{pmatrix} = \begin{pmatrix} -9 \\ 5 \\ -17 \end{pmatrix}$$

So

$$\left| \overrightarrow{AB} \times \overrightarrow{AC} \right| = \sqrt{(-9)^2 + 5^2 + (-17)^2} = \sqrt{81 + 25 + 289} = \sqrt{395}$$

Therefore

$$\text{Area of triangle} = \frac{1}{2}\sqrt{395}$$

So the area is $\dfrac{\sqrt{395}}{2}$ square units, which is approximately 9.94 square units.

10. With respect to a fixed origin O the point A has coordinates $(9, 10, -1)$ and the line ℓ is the intersection of the planes $x - z = -2$ and $x + 2y = 5$.

The points B and C lie on ℓ such that $AB = AC = 15$.

Given that A does not lie on ℓ and that the z coordinate of B is negative

(a) determine the coordinates of B and the coordinates of C. **[4]**

(b) Hence determine a Cartesian equation of the plane containing the points A, B and C. **[3]**

The point D has coordinates $(8, 4, \alpha)$, where α is a constant.

Given that the volume of the tetrahedron $ABCD$ is 156

(c) determine the possible values of α. **[4]**

Given that $\alpha > 0$

(d) determine the shortest distance between the line ℓ and the line passing through the points A and D, giving your answer to 2 significant figures. **[4]**

Solution

(a) Since ℓ is the intersection of

$$x - z = -2 \qquad \text{and} \qquad x + 2y = 5$$

let $z = t$. Then

$$x = t - 2$$

and substituting into $x + 2y = 5$ gives

$$t - 2 + 2y = 5$$

so

$$2y = 7 - t \qquad \Rightarrow \qquad y = \frac{7 - t}{2}$$

Hence a general point P on ℓ is

$$P\left(t - 2, \frac{7 - t}{2}, t\right)$$

Now $AB = AC = 15$, so points B and C on ℓ satisfy $AP = 15$.
Using $A = (9, 10, -1)$,

$$\begin{aligned} AP^2 &= (t - 2 - 9)^2 + \left(\frac{7 - t}{2} - 10\right)^2 + (t - (-1))^2 \\ &= (t - 11)^2 + \left(\frac{-t - 13}{2}\right)^2 + (t + 1)^2 \end{aligned}$$

Since $AP = 15$,

$$(t - 11)^2 + \left(\frac{t + 13}{2}\right)^2 + (t + 1)^2 = 225$$

Multiply by 4:

$$\begin{aligned} 4(t-11)^2 + (t+13)^2 + 4(t+1)^2 &= 900 \\ 4(t^2 - 22t + 121) + (t^2 + 26t + 169) + 4(t^2 + 2t + 1) &= 900 \\ 9t^2 - 54t + 657 &= 900 \\ 9t^2 - 54t - 243 &= 0 \\ t^2 - 6t - 27 &= 0 \\ (t-9)(t+3) &= 0 \end{aligned}$$

So

$$t = 9 \quad \text{or} \quad t = -3$$

If $t = -3$,

$$P = (-5, 5, -3)$$

If $t = 9$,

$$P = (7, -1, 9)$$

Given that the z-coordinate of B is negative, we take

$$B = (-5, 5, -3)$$

and

$$C = (7, -1, 9)$$

Therefore the coordinates are $B = (-5, 5, -3)$ and $C = (7, -1, 9)$.

(b) The plane through A, B and C has normal vector

$$\overrightarrow{AB} \times \overrightarrow{AC}$$

First find the two vectors:

$$\overrightarrow{AB} = \begin{pmatrix} -5-9 \\ 5-10 \\ -3-(-1) \end{pmatrix} = \begin{pmatrix} -14 \\ -5 \\ -2 \end{pmatrix}$$

$$\overrightarrow{AC} = \begin{pmatrix} 7-9 \\ -1-10 \\ 9-(-1) \end{pmatrix} = \begin{pmatrix} -2 \\ -11 \\ 10 \end{pmatrix}$$

Now

$$\begin{aligned} \overrightarrow{AB} \times \overrightarrow{AC} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -14 & -5 & -2 \\ -2 & -11 & 10 \end{vmatrix} \\ &= \begin{pmatrix} -72 \\ 144 \\ 144 \end{pmatrix} \end{aligned}$$

So a normal vector is

$$\begin{pmatrix} -1 \\ 2 \\ 2 \end{pmatrix}$$

Using the point $A = (9, 10, -1)$, the plane equation is

$$-(x-9) + 2(y-10) + 2(z+1) = 0$$

Simplifying,

$$\begin{aligned} -x + 9 + 2y - 20 + 2z + 2 &= 0 \\ -x + 2y + 2z - 9 &= 0 \end{aligned}$$

Hence

$$x - 2y - 2z + 9 = 0$$

(c) From part (b),

$$\overrightarrow{AB} \times \overrightarrow{AC} = \begin{pmatrix} -72 \\ 144 \\ 144 \end{pmatrix}$$

Also

$$D = (8, 4, \alpha)$$

so

$$\overrightarrow{AD} = \begin{pmatrix} 8-9 \\ 4-10 \\ \alpha-(-1) \end{pmatrix} = \begin{pmatrix} -1 \\ -6 \\ \alpha+1 \end{pmatrix}$$

The volume of tetrahedron $ABCD$ is

$$V = \frac{1}{6}\left|(\overrightarrow{AB} \times \overrightarrow{AC}) \cdot \overrightarrow{AD}\right|$$

So

$$\begin{aligned}(\overrightarrow{AB} \times \overrightarrow{AC}) \cdot \overrightarrow{AD} &= \begin{pmatrix} -72 \\ 144 \\ 144 \end{pmatrix} \cdot \begin{pmatrix} -1 \\ -6 \\ \alpha+1 \end{pmatrix} \\ &= 72 - 864 + 144(\alpha+1) \\ &= 72 - 864 + 144\alpha + 144 \\ &= 144\alpha - 648\end{aligned}$$

Given that the volume is 156,

$$\frac{1}{6}|144\alpha - 648| = 156$$

So

$$|144\alpha - 648| = 936$$

Divide by 72:

$$|2\alpha - 9| = 13$$

Hence

$$2\alpha - 9 = 13 \quad \text{or} \quad 2\alpha - 9 = -13$$

giving

$$\alpha = 11 \quad \text{or} \quad \alpha = -2$$

Therefore the possible values of α are 11 and -2.

(d) Since $\alpha > 0$, from part (c) we use

$$\alpha = 11$$

So

$$D = (8, 4, 11)$$

A direction vector for ℓ is obtained from its parametric form:

$$(x, y, z) = \left(t - 2, \frac{7-t}{2}, t\right)$$

so a direction vector is

$$\begin{pmatrix} 1 \\ -\frac{1}{2} \\ 1 \end{pmatrix}$$

and hence we can use

$$\mathbf{d}_1 = \begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix}$$

The line through A and D has direction vector

$$\mathbf{d}_2 = \overrightarrow{AD} = \begin{pmatrix} 8-9 \\ 4-10 \\ 11-(-1) \end{pmatrix} = \begin{pmatrix} -1 \\ -6 \\ 12 \end{pmatrix}$$

Using $B = (-5, 5, -3)$ on ℓ, we have

$$\overrightarrow{BA} = \begin{pmatrix} 9-(-5) \\ 10-5 \\ -1-(-3) \end{pmatrix} = \begin{pmatrix} 14 \\ 5 \\ 2 \end{pmatrix}$$

For two skew lines,

$$\text{shortest distance} = \frac{\left|\overrightarrow{BA} \cdot (\mathbf{d}_1 \times \mathbf{d}_2)\right|}{|\mathbf{d}_1 \times \mathbf{d}_2|}$$

Now find the cross product:

$$\begin{aligned} \mathbf{d}_1 \times \mathbf{d}_2 &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & -1 & 2 \\ -1 & -6 & 12 \end{vmatrix} \\ &= \begin{pmatrix} 0 \\ -26 \\ -13 \end{pmatrix} \end{aligned}$$

Its magnitude is

$$|\mathbf{d}_1 \times \mathbf{d}_2| = \sqrt{0^2 + (-26)^2 + (-13)^2} = \sqrt{845} = 13\sqrt{5}$$

Also

$$\begin{aligned} \overrightarrow{BA} \cdot (\mathbf{d}_1 \times \mathbf{d}_2) &= \begin{pmatrix} 14 \\ 5 \\ 2 \end{pmatrix} \cdot \begin{pmatrix} 0 \\ -26 \\ -13 \end{pmatrix} \\ &= 0 - 130 - 26 \\ &= -156 \end{aligned}$$

So the distance is

$$\begin{aligned} \frac{|-156|}{13\sqrt{5}} &= \frac{156}{13\sqrt{5}} \\ &= \frac{12}{\sqrt{5}} \\ &= \frac{12\sqrt{5}}{5} \end{aligned}$$

and numerically

$$\frac{12\sqrt{5}}{5} \approx 5.37$$

Therefore the shortest distance is

$$\frac{12\sqrt{5}}{5}$$

which is 5.4 to 2 significant figures.